Self Learning

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MySQL Lab Exercises

1. Explain EXISTS vs IN performance comparison with an example.

```
d6_96728_shaurya> use sales;
Database changed
d6_96728_shaurya> SELECT cname
    -> FROM customers
    -> WHERE cnum IN (
    ->     SELECT cnum
    ->     FROM orders
    -> );
+----------+
| cname    |
+----------+
| Hoffman  |
| Giovanni |
| Liu      |
| Grass    |
| Clemens  |
| Cisneros |
| Pereira  |
+----------+
7 rows in set (0.060 sec)

d6_96728_shaurya>
```

```
d6_96728_shaurya> SELECT cname
    -> FROM customers c
    -> WHERE EXISTS (
    ->     SELECT 1
    ->     FROM orders o
    ->     WHERE o.cnum = c.cnum
    -> );
+----------+
| cname    |
+----------+
| Hoffman  |
| Giovanni |
| Liu      |
| Grass    |
| Clemens  |
| Cisneros |
| Pereira  |
+----------+
7 rows in set (0.023 sec)

d6_96728_shaurya>
```

2. Implement a trigger to update the product price in cart, when price is updated in products table

```
d6_96728_shaurya> CREATE TABLE users (
    ->     id INT PRIMARY KEY AUTO_INCREMENT,
    ->     username VARCHAR(50) UNIQUE NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.557 sec)

d6_96728_shaurya>
d6_96728_shaurya> CREATE TABLE products (
    ->     id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(255) NOT NULL,
    ->     price DECIMAL(10,2) NOT NULL,
    ->     stock INT NOT NULL DEFAULT 0
    -> );
Query OK, 0 rows affected (0.256 sec)

d6_96728_shaurya>
d6_96728_shaurya> CREATE TABLE cart (
    ->     id INT PRIMARY KEY AUTO_INCREMENT,
    ->     user_id INT NOT NULL,
    ->     product_id INT NOT NULL,
    ->     price DECIMAL(10,2) NOT NULL,
    ->     quantity INT NOT NULL DEFAULT 1,
    ->     FOREIGN KEY (user_id) REFERENCES users(id),
    ->     FOREIGN KEY (product_id) REFERENCES products(id)
    -> );
Query OK, 0 rows affected (0.545 sec)

d6_96728_shaurya>
```

```
d6_96728_shaurya> DELIMITER //
d6_96728_shaurya>
d6_96728_shaurya> CREATE TRIGGER update_cart_price
    -> AFTER UPDATE ON products
    -> FOR EACH ROW
    -> BEGIN
    ->     -- check if price is actually changed
    ->     IF OLD.price <> NEW.price THEN
    ->
    ->         -- update cart price for that product
    ->         UPDATE cart
    ->         SET price = NEW.price
    ->         WHERE product_id = NEW.id;
    ->
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.121 sec)

d6_96728_shaurya>
d6_96728_shaurya> DELIMITER ;
d6_96728_shaurya>
```

```
d6_96728_shaurya> INSERT INTO users (username, email)
    -> VALUES ('Amit', 'amit@mail.com');
Query OK, 1 row affected (0.129 sec)

d6_96728_shaurya>
d6_96728_shaurya> INSERT INTO products (name, price, stock)
    -> VALUES ('Laptop', 50000, 10);
Query OK, 1 row affected (0.063 sec)

d6_96728_shaurya>
d6_96728_shaurya> INSERT INTO cart (user_id, product_id, price, quantity)
    -> VALUES (1, 1, 50000, 1);
Query OK, 1 row affected (0.067 sec)

d6_96728_shaurya>
d6_96728_shaurya> UPDATE products
    -> SET price = 55000
    -> WHERE id = 1;
Query OK, 1 row affected (0.062 sec)
Rows matched: 1  Changed: 1  Warnings: 0

d6_96728_shaurya>
d6_96728_shaurya> SELECT * FROM cart;
+----+---------+------------+----------+----------+
| id | user_id | product_id | price    | quantity |
+----+---------+------------+----------+----------+
|  1 |       1 |          1 | 55000.00 |        1 |
+----+---------+------------+----------+----------+
1 row in set (0.005 sec)

d6_96728_shaurya>
```

3. Understand ON DELETE CASCADE, ON UPDATE CASCADE. Write emp, dept CREATE TABLE statements and apply CASCADE for all foreign keys.

```
d6_96728_shaurya> CREATE TABLE dept (
    ->     deptno INT PRIMARY KEY,
    ->     dname VARCHAR(50),
    ->     loc VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.800 sec)

d6_96728_shaurya>
d6_96728_shaurya> CREATE TABLE emp (
    ->     empno INT PRIMARY KEY,
    ->     ename VARCHAR(50),
    ->     job VARCHAR(50),
    ->     sal DECIMAL(10,2),
    ->     deptno INT,
    ->
    ->     FOREIGN KEY (deptno)
    ->     REFERENCES dept(deptno)
    ->     ON DELETE CASCADE
    ->     ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.707 sec)
```

```
d6_96728_shaurya>
d6_96728_shaurya> INSERT INTO dept VALUES (10, 'ACCOUNTING', 'PUNE');
Query OK, 1 row affected (0.327 sec)

d6_96728_shaurya> INSERT INTO dept VALUES (20, 'SALES', 'MUMBAI');
Query OK, 1 row affected (0.051 sec)

d6_96728_shaurya>
d6_96728_shaurya> INSERT INTO emp VALUES (1, 'Amit', 'CLERK', 1200, 10);
Query OK, 1 row affected (0.065 sec)

d6_96728_shaurya> INSERT INTO emp VALUES (2, 'Priya', 'MANAGER', 3000, 10);
Query OK, 1 row affected (0.067 sec)

d6_96728_shaurya> INSERT INTO emp VALUES (3, 'Rahul', 'CLERK', 1500, 20);
Query OK, 1 row affected (0.057 sec)
```

```
d6_96728_shaurya> DELETE FROM dept
    -> WHERE deptno = 10;
Query OK, 1 row affected (0.097 sec)

d6_96728_shaurya>
d6_96728_shaurya> SELECT * FROM emp;
+-------+-------+-------+---------+--------+
| empno | ename | job   | sal     | deptno |
+-------+-------+-------+---------+--------+
|     3 | Rahul | CLERK | 1500.00 |     20 |
+-------+-------+-------+---------+--------+
1 row in set (0.003 sec)

d6_96728_shaurya>
d6_96728_shaurya> UPDATE dept
    -> SET deptno = 30
    -> WHERE deptno = 20;
Query OK, 1 row affected (0.057 sec)
Rows matched: 1  Changed: 1  Warnings: 0

d6_96728_shaurya>
d6_96728_shaurya> SELECT * FROM emp;
+-------+-------+-------+---------+--------+
| empno | ename | job   | sal     | deptno |
+-------+-------+-------+---------+--------+
|     3 | Rahul | CLERK | 1500.00 |     30 |
+-------+-------+-------+---------+--------+
1 row in set (0.004 sec)

d6_96728_shaurya> UPDATE dept
    -> SET deptno = 30
    -> WHERE deptno = 20;
Query OK, 0 rows affected (0.005 sec)
Rows matched: 0  Changed: 0  Warnings: 0

d6_96728_shaurya>
```

MongoDb Lab

Exercises 1. Practice CRUD operations on Mongo.

1. Create two collections -- emp and dept.

```
selfstudy> db.createCollection("emp");
{ ok: 1 }
selfstudy> db.createCollection("dept");
{ ok: 1 }
selfstudy>
```

2. Insert the emp and dept records.

```
selfstudy> db.dept.insertMany([
...   { deptno: 10, dname: "ACCOUNTING", loc: "PUNE" },
...   { deptno: 20, dname: "RESEARCH", loc: "MUMBAI" },
...   { deptno: 30, dname: "SALES", loc: "DELHI" },
...   { deptno: 40, dname: "HR", loc: "CHENNAI" }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ba95ce6422d2b73546b799'),
    '1': ObjectId('69ba95ce6422d2b73546b79a'),
    '2': ObjectId('69ba95ce6422d2b73546b79b'),
    '3': ObjectId('69ba95ce6422d2b73546b79c')
  }
}
selfstudy> db.emp.insertMany([
...   { ename: "Amit", job: "CLERK", sal: 1200, comm: 100, deptno: 10 },
...   { ename: "Priya", job: "MANAGER", sal: 3000, comm: 500, deptno: 10 },
...   { ename: "Rahul", job: "CLERK", sal: 800, comm: 50, deptno: 20 },
...   { ename: "Sneha", job: "ANALYST", sal: 2500, comm: 300, deptno: 20 },
...   { ename: "Vikram", job: "SALESMAN", sal: 1500, comm: 200, deptno: 30 },
...   { ename: "Anjali", job: "PRESIDENT", sal: 5000, comm: 0, deptno: 10 }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ba95de6422d2b73546b79d'),
    '1': ObjectId('69ba95de6422d2b73546b79e'),
    '2': ObjectId('69ba95de6422d2b73546b79f'),
    '3': ObjectId('69ba95de6422d2b73546b7a0'),
    '4': ObjectId('69ba95de6422d2b73546b7a1'),
    '5': ObjectId('69ba95de6422d2b73546b7a2')
  }
}
```

3. Increase salary of dept 10 emps by 100.

```
selfstudy> db.emp.updateMany(
...   { deptno: 10 },
...   { $inc: { sal: 100 } }
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

4. Display emp name, job, deptno, sal in sorted order of deptno, job, sal (desc) for all emps whose sal more than 1000.

```
selfstudy> db.emp.find(
...   { sal: { $gt: 1000 } },
...   { _id: 0, ename: 1, job: 1, deptno: 1, sal: 1 }
... ).sort({ deptno: 1, job: 1, sal: -1 });
[
  { ename: 'Amit', job: 'CLERK', sal: 1300, deptno: 10 },
  { ename: 'Priya', job: 'MANAGER', sal: 3100, deptno: 10 },
  { ename: 'Anjali', job: 'PRESIDENT', sal: 5100, deptno: 10 },
  { ename: 'Sneha', job: 'ANALYST', sal: 2500, deptno: 20 },
  { ename: 'Vikram', job: 'SALESMAN', sal: 1500, deptno: 30 }
]
```

5. Delete the dept with id 40.

```
selfstudy> db.dept.deleteOne({ deptno: 40 });
{ acknowledged: true, deletedCount: 1 }
```

2. Learn grouping operations using Mongo aggregation pipeline

1. Display job wise total sal, max sal, min sal, and avg sal for all jobs other than PRESIDENT.

```
selfstudy> db.emp.aggregate([
...   { $match: { job: { $ne: "PRESIDENT" } } },
...
...   {
...     $group: {
...       _id: "$job",
...       total_sal: { $sum: "$sal" },
...       max_sal: { $max: "$sal" },
...       min_sal: { $min: "$sal" },
...       avg_sal: { $avg: "$sal" }
...     }
...   }
... ]);
[
  {
    _id: 'ANALYST',
    total_sal: 2500,
    max_sal: 2500,
    min_sal: 2500,
    avg_sal: 2500
  },
  {
    _id: 'SALESMAN',
    total_sal: 1500,
    max_sal: 1500,
    min_sal: 1500,
    avg_sal: 1500
  },
  {
    _id: 'CLERK',
    total_sal: 2100,
    max_sal: 1300,
    min_sal: 800,
    avg_sal: 1050
  },
  {
    _id: 'MANAGER',
    total_sal: 3100,
    max_sal: 3100,
    min_sal: 3100,
    avg_sal: 3100
  }
]
```

2. Display emp name, dept name, income (sal + comm) in sorted order of their income.

```
selfstudy> db.emp.aggregate([
...   {
...     $lookup: {
...       from: "dept",
...       localField: "deptno",
...       foreignField: "deptno",
...       as: "dept_info"
...     }
...   },
...
...   { $unwind: "$dept_info" },
...
...   {
...     $addFields: {
...       income: { $add: ["$sal", "$comm"] }
...     }
...   },
...
...   {
...     $project: {
...       _id: 0,
...       ename: 1,
...       dept_name: "$dept_info.dname",
...       income: 1
...     }
...   },
...
...   { $sort: { income: -1 } }
... ]);
[
  { ename: 'Anjali', income: 5100, dept_name: 'ACCOUNTING' },
  { ename: 'Priya', income: 3600, dept_name: 'ACCOUNTING' },
  { ename: 'Sneha', income: 2800, dept_name: 'RESEARCH' },
  { ename: 'Vikram', income: 1700, dept_name: 'SALES' },
  { ename: 'Amit', income: 1400, dept_name: 'ACCOUNTING' },
  { ename: 'Rahul', income: 850, dept_name: 'RESEARCH' }
]
```